36-671 Final Project

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Research question: How do regional gasoline prices co-move over time, and what are the lead-lag relationships between regions and between crude oil prices and retail gasoline prices?

Finding, downloading and cleaning up relevant time series data

```r
# Load weekly gasoline data
# skip = 2 because row 1 = title, row 2 = source key, row 3 = actual variable names
us_data <- read_excel(file_path, sheet = "Data 1", skip = 2) |>
  rename(Date = 1, US_All_Grades = 2) |>
  mutate(Date = as.Date(Date), US_All_Grades = as.numeric(US_All_Grades)) |>
  arrange(Date)

region_data <- read_excel(file_path, sheet = "Data 2", skip = 2) |>
  rename(
    Date = 1,
    East_Coast = 2,
    New_England = 3,
    Central_Atlantic = 4,
    Lower_Atlantic = 5,
    Midwest = 6,
    Gulf_Coast = 7,
    Rocky_Mountain = 8,
    West_Coast = 9,
    West_Coast_ex_CA = 10
  ) |>
  mutate(Date = as.Date(Date)) |>
  mutate(across(-Date, as.numeric)) |>
  arrange(Date)

# Join national + five regional series, drop any incomplete weeks
gas_complete <- us_data |>
  left_join(
    region_data |> dplyr::select(Date, East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast),
    by = "Date"
  ) |>
  drop_na()

cat("Weekly gasoline date range:", min(gas_complete$Date), "to", max(gas_complete$Date), "\n")
```

```
## Weekly gasoline date range: 8495 to 20542
```

```r
cat("Observations:", nrow(gas_complete), "\n")
```

```
## Observations: 1722
```

```r
# Load weekly Brent crude oil data
crude_data <- read_csv(brent_path) |>
  rename(Brent = Brent_Price_USD) |>
  mutate(Date = as.Date(Date), Brent = as.numeric(Brent)) |>
  arrange(Date)

cat("Crude oil date range:", min(crude_data$Date), "to", max(crude_data$Date), "\n")
```

```
## Crude oil date range: 8404 to 20535
```

```r
# Convert all series to weekly ts objects
# Get week number via %U (Sunday start); +1 to make it start from 1
start_vec <- c(
  as.numeric(format(min(gas_complete$Date), "%Y")),
  as.numeric(format(min(gas_complete$Date), "%U")) + 1
)

us_ts <- ts(gas_complete$US_All_Grades, start = start_vec, frequency = 52)
east_ts <- ts(gas_complete$East_Coast, start = start_vec, frequency = 52)
midwest_ts <- ts(gas_complete$Midwest, start = start_vec, frequency = 52)
gulf_ts <- ts(gas_complete$Gulf_Coast, start = start_vec, frequency = 52)
rocky_ts <- ts(gas_complete$Rocky_Mountain, start = start_vec, frequency = 52)
west_ts <- ts(gas_complete$West_Coast, start = start_vec, frequency = 52)

# Build multivariate ts for VAR analysis
regions_ts <- gas_complete |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  as.matrix() |>
  ts(start = start_vec, frequency = 52)

colnames(regions_ts) <- c("East_Coast", "Midwest", "Gulf_Coast", "Rocky_Mountain", "West_Coast")
```

```r
# Both series are weekly, join directly on date
gas_crude <- gas_complete |>
  inner_join(crude_data, by = "Date") |>
  drop_na()

cat("Gas price-crude oil overlapping date range:", min(gas_crude$Date), "to", max(gas_crude$Date), "\n")
```

```
## Gas price-crude oil overlapping date range: 8495 to 20535
```

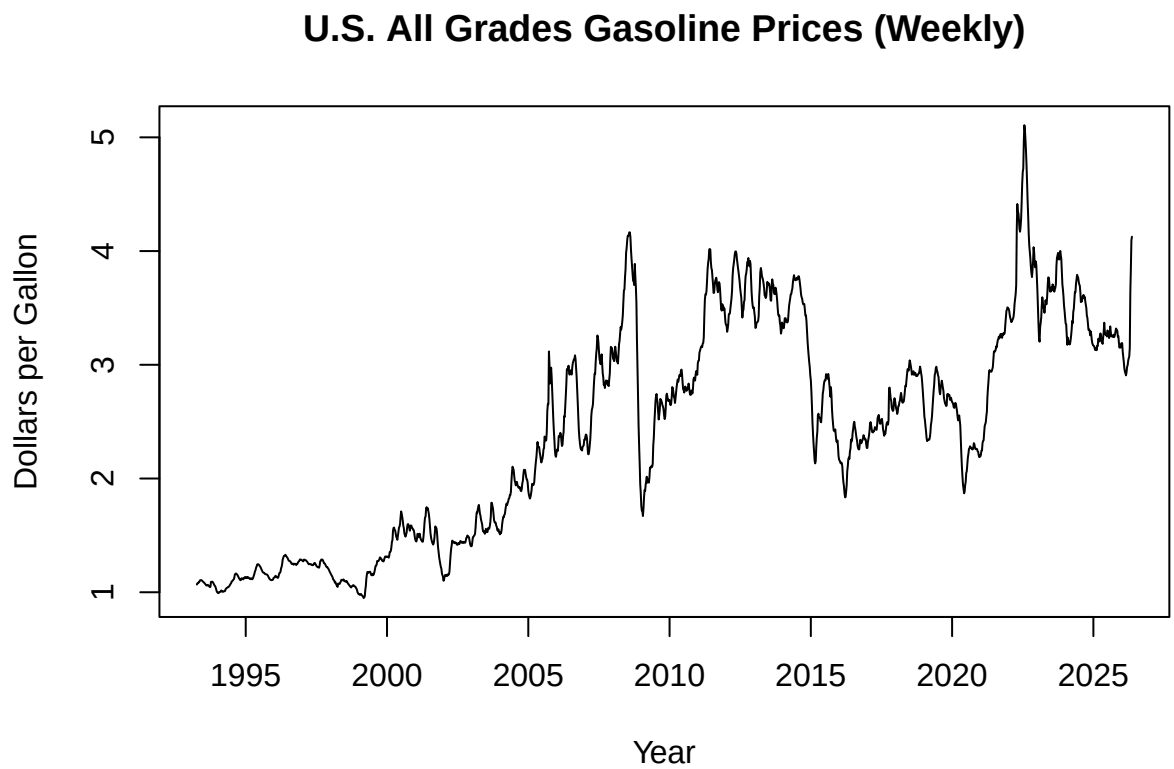
```r
cat("Observations:", nrow(gas_crude), "\n")
```

```
## Observations: 1721
```

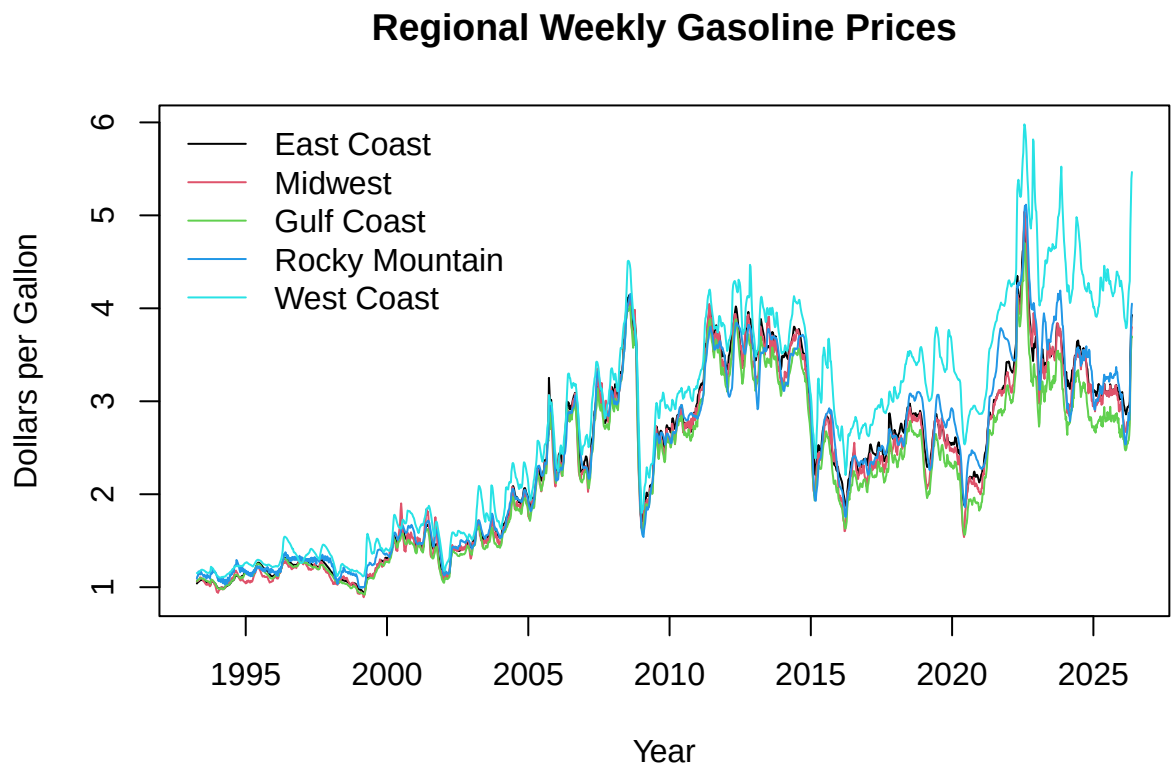
```r
# Shared start for weekly ts
start_crude <- c(
  as.numeric(format(min(gas_crude$Date), "%Y")),
  as.numeric(format(min(gas_crude$Date), "%U")) + 1
)

gas_ts_crude <- ts(gas_crude$US_All_Grades, start = start_crude, frequency = 52)
crude_ts <- ts(gas_crude$Brent, start = start_crude, frequency = 52)
```

Exploratory Data Analysis

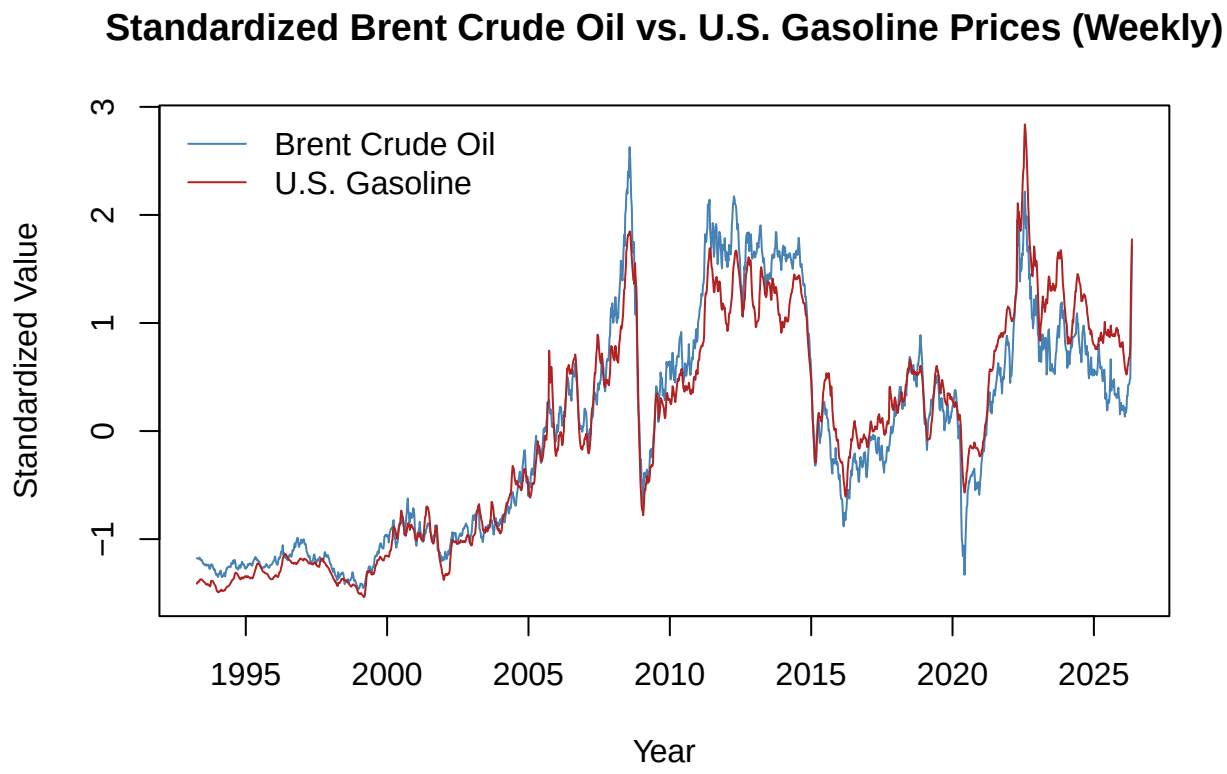
```
# Time series plots: national, regional, and crude oil
plot(us_ts,
     main = "U.S. All Grades Gasoline Prices (Weekly)",
     xlab = "Year", ylab = "Dollars per Gallon")
```




```
ts.plot(east_ts, midwest_ts, gulf_ts, rocky_ts, west_ts,
        col = 1:5, lty = 1,
        main = "Regional Weekly Gasoline Prices",
        xlab = "Year", ylab = "Dollars per Gallon")
legend("topleft",
       legend = c("East Coast", "Midwest", "Gulf Coast", "Rocky Mountain", "West Coast"),
       col = 1:5, lty = 1, bty = "n")
```




```r
# Standardize for comparison since crude oil and gas prices are on diff scales/units
crude_scaled <- ts(
  as.numeric(scale(as.numeric(crude_ts))),
  start = start(crude_ts),
  frequency = frequency(crude_ts)
)

gas_scaled <- ts(
  as.numeric(scale(as.numeric(gas_ts_crude))),
  start = start(gas_ts_crude),
  frequency = frequency(gas_ts_crude)
)

ts.plot(crude_scaled, gas_scaled,
        col = c("steelblue", "firebrick"), lty = 1,
        main = "Standardized Brent Crude Oil vs. U.S. Gasoline Prices (Weekly)",
        xlab = "Year", ylab = "Standardized Value")
legend("topleft",
       legend = c("Brent Crude Oil", "U.S. Gasoline"),
       col = c("steelblue", "firebrick"), lty = 1, bty = "n")
```



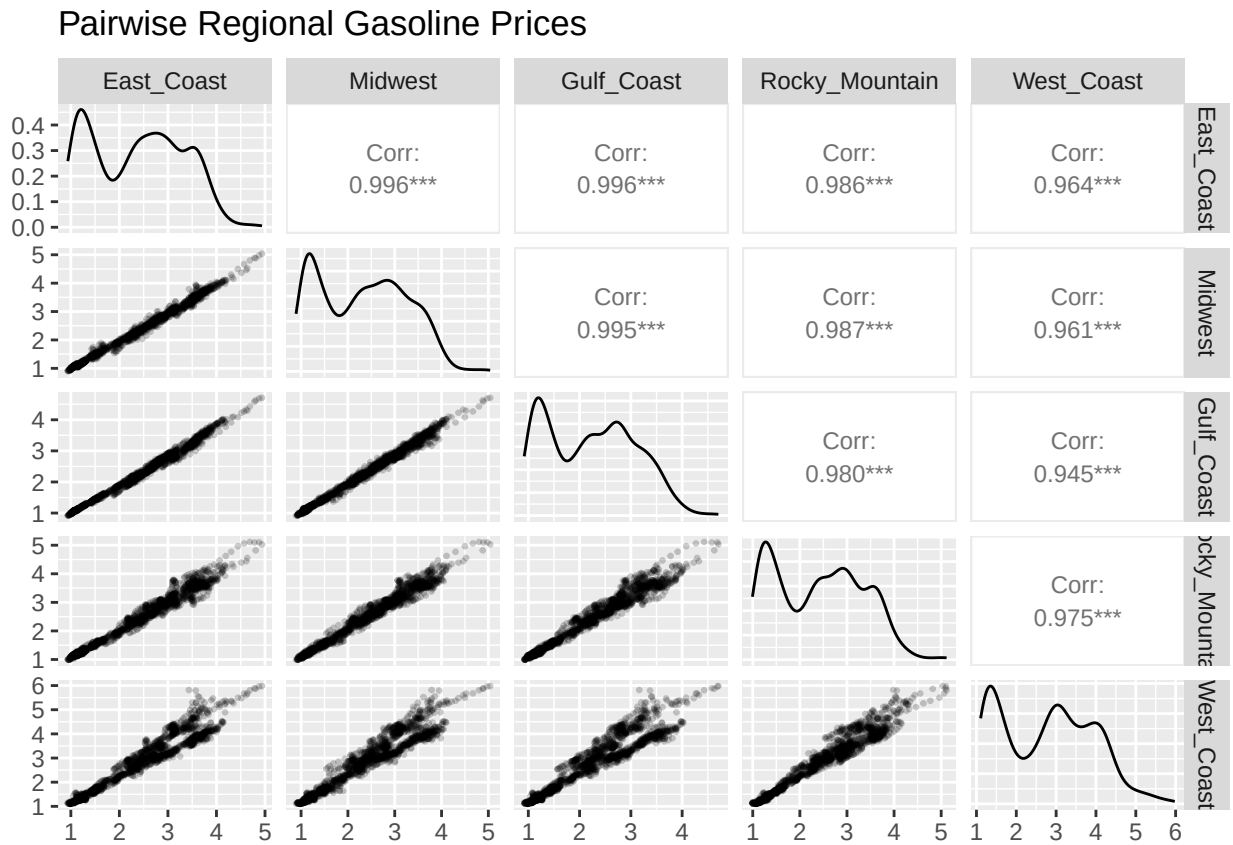
The national gasoline series shows long swings over time rather than fluctuating around a constant level, suggesting strong persistence and possible nonstationarity in the level series. There seems to be a general upward movement over the sample. The plots also suggest changing variability over time, with larger swings during some periods, especially around major shocks, which may indicate heteroskedasticity.

The regional plot shows that all five regions follow the same broad pattern, rising and falling during the same major episodes.

At the same time, the regions are not exactly the same. The West Coast is generally higher than the other regions and appears more volatile, especially in later years. Rocky Mountain also shows somewhat greater variability at certain points, while the Gulf Coast and Midwest tend to track the national pattern more closely and at lower price levels. So the data show both strong shared movement and meaningful regional differences.

Comparing Brent crude to U.S. gasoline, the two series move very closely together over time. This suggests that crude oil is an important driver of gasoline prices, although there are other factors like refining, transportation, taxes, regional conditions, etc.

```r
# Pairwise scatterplots to check contemporaneous correlations before looking at lags
gas_complete |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  ggpairs(
    title = "Pairwise Regional Gasoline Prices",
    lower = list(continuous = wrap("points", alpha = 0.2, size = 0.5)),
    upper = list(continuous = wrap("cor", size = 3))
  )
```

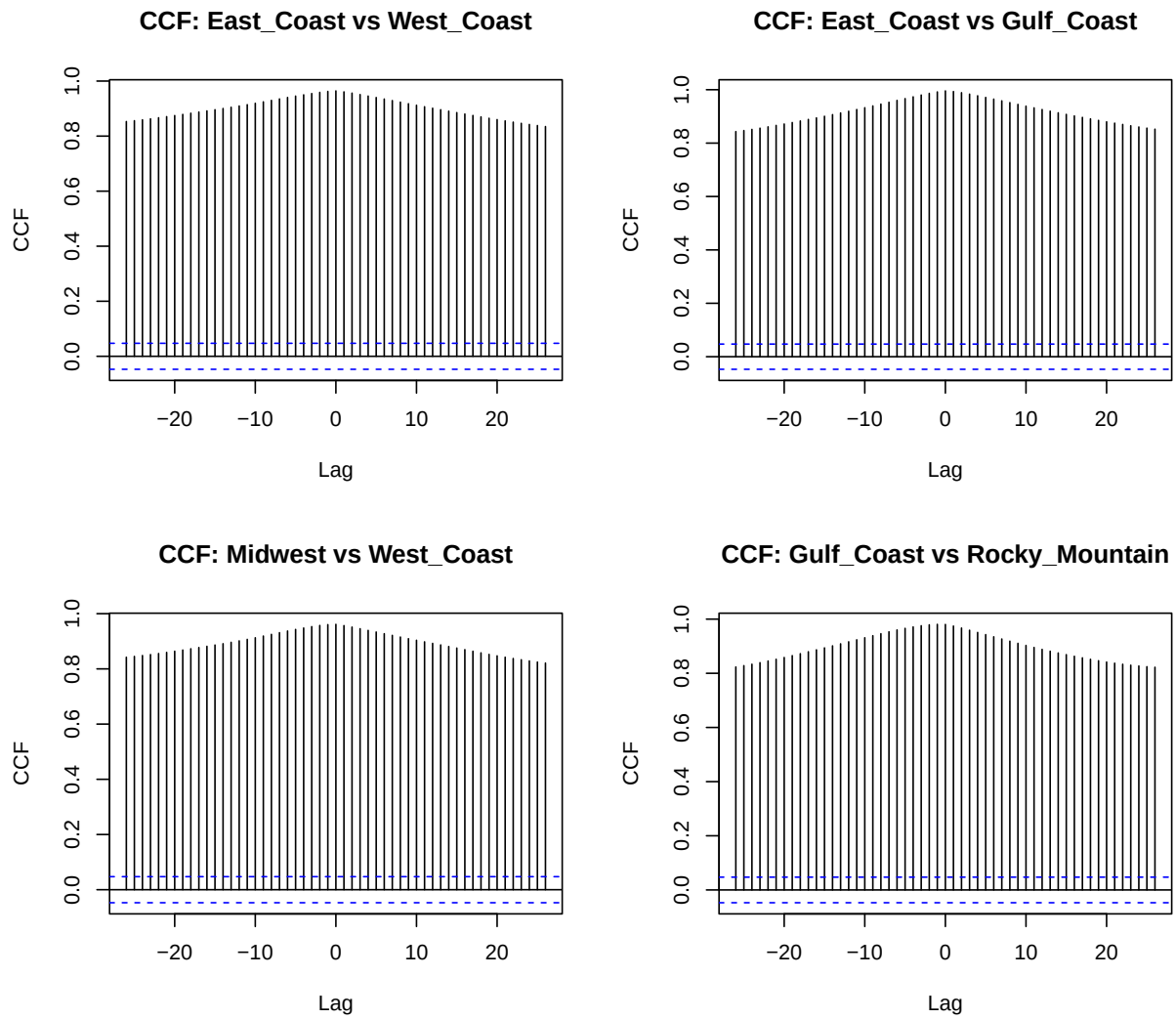


The pairwise scatterplots indicate strong positive contemporaneous relationships among weekly gasoline prices across all regions. For every regional pair, the points form a clear upward-sloping pattern, suggesting that when prices are relatively high in one region, they also tend to be relatively high in the other during the same week. This is consistent with the very high pairwise correlations, which range from 0.945 to 0.996. Several pairs, such as East Coast and Midwest, fall especially tightly along a line, indicating very strong same-week linear association. Pairs involving the West Coast appear somewhat more dispersed by comparison, though the relationship remains strongly positive. However, because these plots compare observations from the same week only, they do not provide evidence about lead-lag structure or whether price changes in one region tend to precede changes in another.

This shows that the regions are clearly not independent; therefore it is reasonable to study them as a connected multivariate time series. Having established strong contemporaneous correlation, we can next investigate whether some regions tend to lead or lag others over time. This provides a natural motivation for using lag-based analysis and vector autoregressive models.

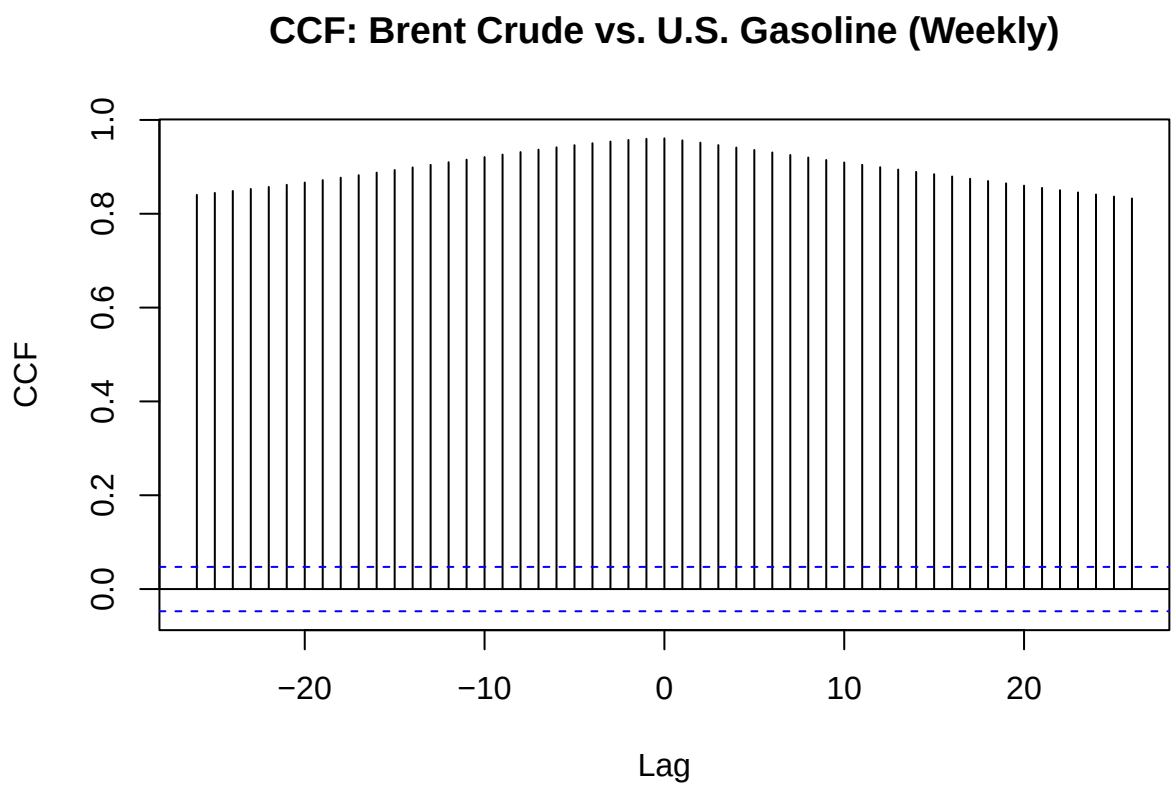
```r
# Cross-correlation between region pairs
# negative lag k means x leads y by k weeks
region_pairs <- list(
  c("East_Coast", "West_Coast"),
  c("East_Coast", "Gulf_Coast"),
  c("Midwest", "West_Coast"),
  c("Gulf_Coast", "Rocky_Mountain")
)

par(mfrow = c(2, 2))
for (pair in region_pairs) {
  ccf(gas_complete[[pair[1]]], gas_complete[[pair[2]]],
```

```r
      main = paste("CCF:", pair[1], "vs", pair[2]),
      lag.max = 26,  # up to ~6 months of weekly lags
      ylab = "CCF")
}
```




```r
par(mfrow = c(1, 1))
```

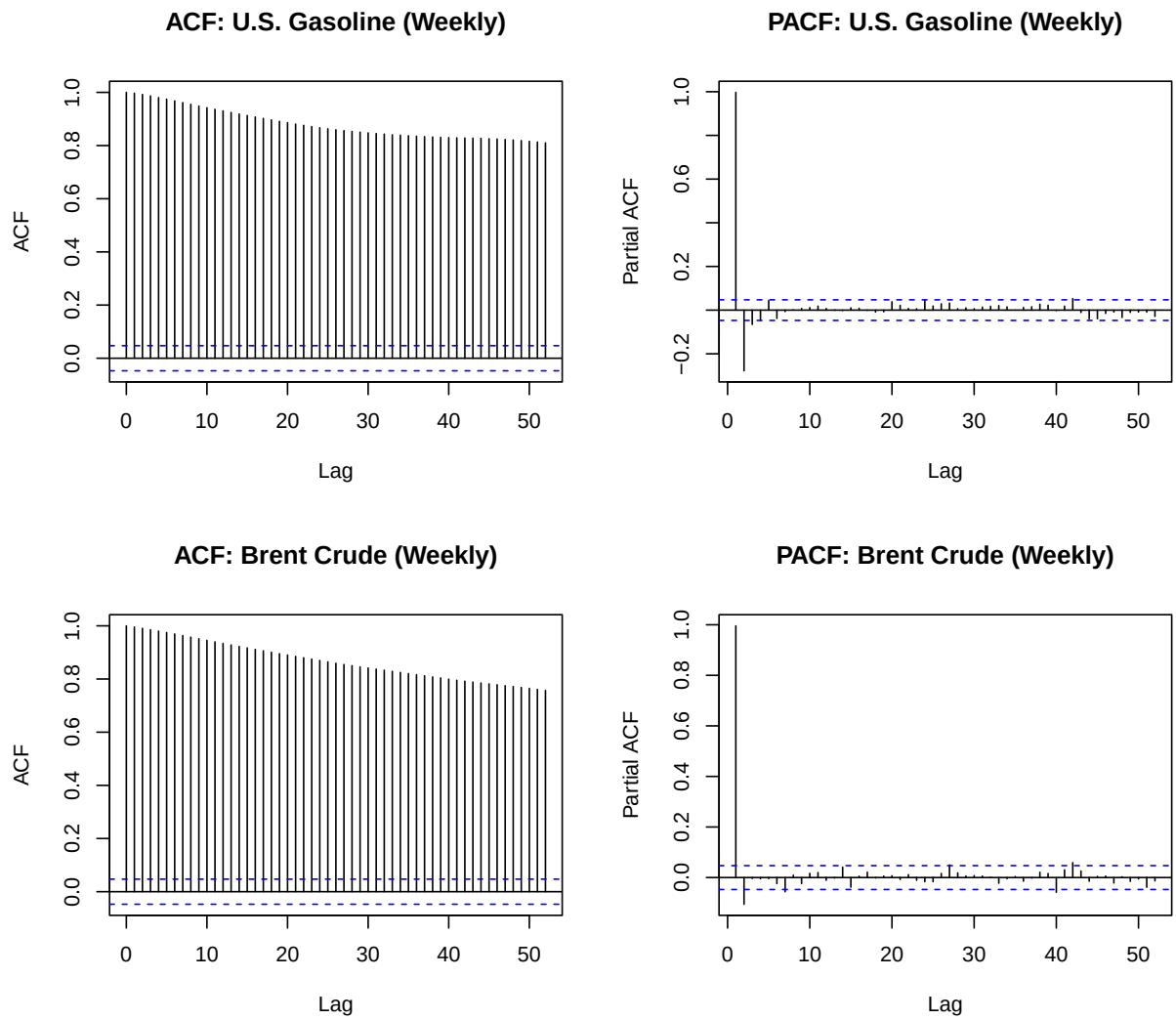
```r
# Cross-correlation: Brent vs. national gasoline
# negative lags = Brent leads gas, positive lags = gas leads Brent
ccf(gas_crude$Brent, gas_crude$US_All_Grades,
    main = "CCF: Brent Crude vs. U.S. Gasoline (Weekly)",
    lag.max = 26,
    ylab = "CCF")
```



The cross-correlation plots show that the regional gasoline price series are strongly related not just in the same week, but across many nearby lags as well. The highest correlations occur around lag 0, which points to strong contemporaneous movement (consistent with results from pairwise scatterplots above). At the same time, the correlations stay very high across the full lag window, so the raw series do not show a clear lead-lag pattern where one region consistently moves first and another follows after a specific number of weeks. The same pattern appears for Brent crude and U.S. gasoline prices, suggesting strong overall co-movement but not a clean lag interpretation in levels.

So we definitely need to check stationarity and do lead-lag interpretation on transformed stationary series rather than raw data.

```
# ACF and PACF for national gasoline and Brent to characterize autocorrelation before fitting VAR
par(mfrow = c(2, 2))
acf(gas_complete$US_All_Grades, main = "ACF: U.S. Gasoline (Weekly)", lag.max = 52)
pacf(gas_complete$US_All_Grades, main = "PACF: U.S. Gasoline (Weekly)", lag.max = 52)
acf(gas_crude$Brent, main = "ACF: Brent Crude (Weekly)", lag.max = 52)
pacf(gas_crude$Brent, main = "PACF: Brent Crude (Weekly)", lag.max = 52)
```




```r
par(mfrow = c(1, 1))
```

The ACF and PACF plots show that both weekly U.S. gasoline and Brent crude prices are highly persistent over time. The ACF stays large and declines slowly, which suggests that prices are strongly related to their past values even many weeks back. The PACF has 2 significant spikes at lags 1 and 2 and cuts off after, which suggests AR structure? Doesn't looks like there is clear seasonality over the first year of weekly lags.

Model Selection - Regional VAR Model

```r
# log for variation
gas_complete$East_Coast <- log(gas_complete$East_Coast)
gas_complete$Midwest <- log(gas_complete$Midwest)
gas_complete$Gulf_Coast <- log(gas_complete$Gulf_Coast)
gas_complete$Rocky_Mountain <- log(gas_complete$Rocky_Mountain)
gas_complete$West_Coast <- log(gas_complete$West_Coast)
# subtract mean to detrend
gas_complete$East_Coast <- gas_complete$East_Coast - mean(gas_complete$East_Coast)
gas_complete$Midwest <- gas_complete$Midwest - mean(gas_complete$Midwest)
```

```r
gas_complete$Gulf_Coast <- gas_complete$Gulf_Coast - mean(gas_complete$Gulf_Coast)
gas_complete$Rocky_Mountain <- gas_complete$Rocky_Mountain - mean(gas_complete$Rocky_Mountain)
gas_complete$West_Coast <- gas_complete$West_Coast - mean(gas_complete$West_Coast)
```

```r
x = ts(dplyr::select(gas_complete, East_Coast:West_Coast),
       # fourteenth week of 1993
       start = c(1993, 14), frequency = 52)
VARselect(x, lag.max = 14, type = 'none')
```

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      4      4      2      4
##
## $criteria
##                    1             2             3             4             5
## AIC(n) -4.330373e+01 -4.414999e+01 -4.420712e+01 -4.424607e+01 -4.424050e+01
## HQ(n)  -4.327424e+01 -4.409101e+01 -4.411866e+01 -4.412812e+01 -4.409306e+01
## SC(n)  -4.322406e+01 -4.399065e+01 -4.396811e+01 -4.392739e+01 -4.384215e+01
## FPE(n)  1.561097e-19  6.697379e-20  6.325471e-20  6.083865e-20  6.117874e-20
##                    6             7             8             9            10
## AIC(n) -4.423548e+01 -4.422686e+01 -4.421603e+01 -4.422077e+01 -4.421300e+01
## HQ(n)  -4.405856e+01 -4.402045e+01 -4.398014e+01 -4.395538e+01 -4.391813e+01
## SC(n)  -4.375746e+01 -4.366917e+01 -4.357867e+01 -4.350373e+01 -4.341630e+01
## FPE(n)  6.148696e-20  6.201965e-20  6.269590e-20  6.240109e-20  6.288878e-20
##                   11            12            13            14
## AIC(n) -4.419802e+01 -4.419093e+01 -4.418949e+01 -4.417450e+01
## HQ(n)  -4.387366e+01 -4.383708e+01 -4.380616e+01 -4.376168e+01
## SC(n)  -4.332164e+01 -4.323488e+01 -4.315378e+01 -4.305911e+01
## FPE(n)  6.384006e-20  6.429645e-20  6.439118e-20  6.536712e-20
```

Using p = 2 due to EDA ACF/PACF graphs + SC (same as BIC).

```r
fit_regional = VAR(x, p = 2, type = 'none')
summary(fit_regional)
```

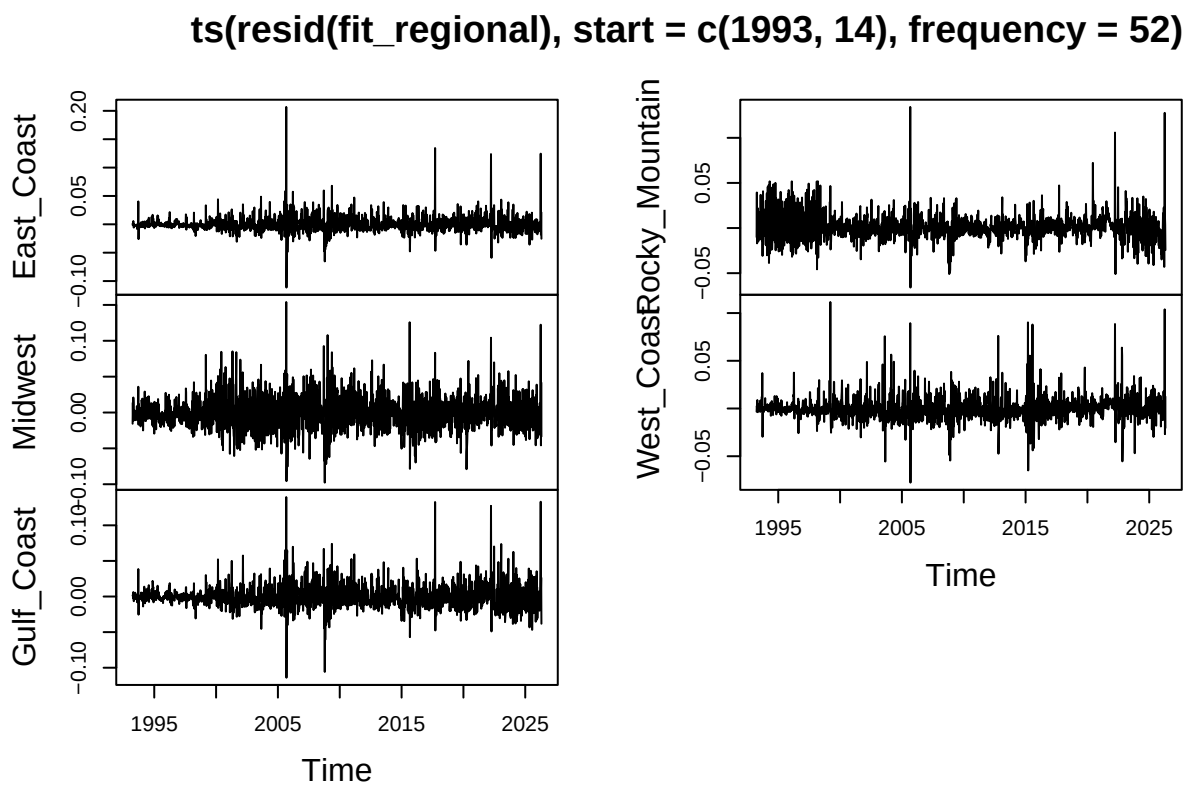
```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast
## Deterministic variables: none
## Sample size: 1720
## Log Likelihood: 25831.5
## Roots of the characteristic polynomial:
## 0.9971 0.977 0.9177 0.9177 0.7807 0.6594 0.3422 0.2416 0.2416 0.1496
## Call:
## VAR(y = x, p = 2, type = "none")
##
##
## Estimation results for equation East_Coast:
## ===========================================
## East_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + East_Co
##
##                    Estimate Std. Error t value Pr(>|t|)
## East_Coast.l1      0.977576   0.053067  18.421  < 2e-16 ***
## Midwest.l1         0.222695   0.021587  10.316  < 2e-16 ***
```

```
## Gulf_Coast.l1      0.203249   0.045365   4.480 7.95e-06 ***
## Rocky_Mountain.l1 -0.006981   0.027587  -0.253 0.800241
## West_Coast.l1      0.107180   0.027497   3.898 0.000101 ***
## East_Coast.l2     -0.022282   0.052392  -0.425 0.670673
## Midwest.l2        -0.173130   0.021438  -8.076 1.25e-15 ***
## Gulf_Coast.l2     -0.191902   0.044926  -4.272 2.05e-05 ***
## Rocky_Mountain.l2 -0.031568   0.026329  -1.199 0.230700
## West_Coast.l2     -0.087773   0.028078  -3.126 0.001801 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.01455 on 1710 degrees of freedom
## Multiple R-Squared: 0.9989,  Adjusted R-squared: 0.9989
## F-statistic: 1.526e+05 on 10 and 1710 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation Midwest:
## ========================================
## Midwest = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + East_Coas
##
##                    Estimate Std. Error t value Pr(>|t|)
## East_Coast.l1      -0.23605    0.08929  -2.644 0.008275 **
## Midwest.l1          1.25762    0.03632  34.626  < 2e-16 ***
## Gulf_Coast.l1       0.26166    0.07633   3.428 0.000622 ***
## Rocky_Mountain.l1   0.12366    0.04641   2.664 0.007788 **
## West_Coast.l1       0.05610    0.04626   1.213 0.225482
## East_Coast.l2       0.29233    0.08815   3.316 0.000932 ***
## Midwest.l2         -0.36996    0.03607 -10.257  < 2e-16 ***
## Gulf_Coast.l2      -0.20368    0.07559  -2.695 0.007115 **
## Rocky_Mountain.l2 -0.15710    0.04430  -3.546 0.000401 ***
## West_Coast.l2      -0.02719    0.04724  -0.576 0.564994
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.02448 on 1710 degrees of freedom
## Multiple R-Squared: 0.9968,  Adjusted R-squared: 0.9968
## F-statistic: 5.369e+04 on 10 and 1710 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation Gulf_Coast:
## ===========================================
## Gulf_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + East_C
##
##                    Estimate Std. Error t value Pr(>|t|)
## East_Coast.l1       0.04934    0.06392   0.772 0.440251
## Midwest.l1          0.32602    0.02600  12.539  < 2e-16 ***
## Gulf_Coast.l1       1.01806    0.05464  18.632  < 2e-16 ***
## Rocky_Mountain.l1   0.03996    0.03323   1.203 0.229269
## West_Coast.l1       0.11058    0.03312   3.339 0.000859 ***
## East_Coast.l2      -0.02460    0.06311  -0.390 0.696683
## Midwest.l2         -0.27396    0.02582 -10.610  < 2e-16 ***
## Gulf_Coast.l2      -0.06414    0.05411  -1.185 0.236052
```

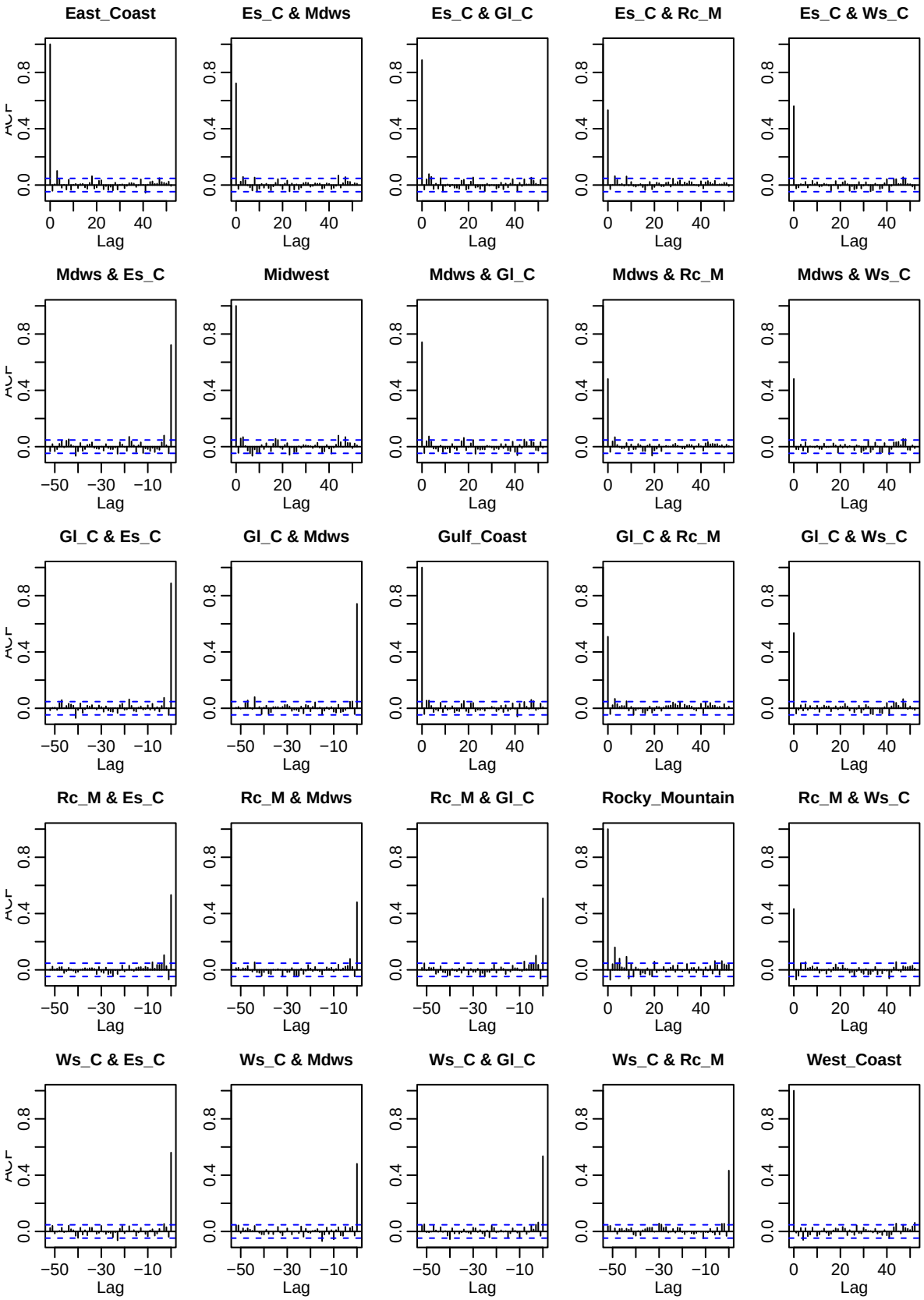
```
## Rocky_Mountain.l2 -0.08519    0.03171  -2.686 0.007294 **
## West_Coast.l2     -0.10205    0.03382  -3.018 0.002585 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.01752 on 1710 degrees of freedom
## Multiple R-Squared: 0.9982,  Adjusted R-squared: 0.9982
## F-statistic: 9.571e+04 on 10 and 1710 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation Rocky_Mountain:
## ===============================================
## Rocky_Mountain = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + Eas
##
##                    Estimate Std. Error t value Pr(>|t|)
## East_Coast.l1     -0.23271    0.05480  -4.247 2.28e-05 ***
## Midwest.l1         0.20631    0.02229   9.256  < 2e-16 ***
## Gulf_Coast.l1      0.21975    0.04684   4.691 2.93e-06 ***
## Rocky_Mountain.l1  1.11376    0.02849  39.100  < 2e-16 ***
## West_Coast.l1      0.25967    0.02839   9.146  < 2e-16 ***
## East_Coast.l2      0.14052    0.05410   2.597  0.00947 **
## Midwest.l2        -0.13803    0.02214  -6.236 5.66e-10 ***
## Gulf_Coast.l2     -0.14082    0.04639  -3.036  0.00244 **
## Rocky_Mountain.l2 -0.22244    0.02719  -8.182 5.39e-16 ***
## West_Coast.l2     -0.20869    0.02899  -7.198 9.10e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.01502 on 1710 degrees of freedom
## Multiple R-Squared: 0.9987,  Adjusted R-squared: 0.9987
## F-statistic: 1.34e+05 on 10 and 1710 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation West_Coast:
## ===========================================
## West_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + East_Co
##
##                    Estimate Std. Error t value Pr(>|t|)
## East_Coast.l1     -0.12517    0.04840  -2.586 0.009785 **
## Midwest.l1         0.15510    0.01969   7.878 5.85e-15 ***
## Gulf_Coast.l1      0.06146    0.04137   1.486 0.137579
## Rocky_Mountain.l1  0.06144    0.02516   2.442 0.014707 *
## West_Coast.l1      1.54850    0.02508  61.750  < 2e-16 ***
## East_Coast.l2      0.17344    0.04778   3.630 0.000292 ***
## Midwest.l2        -0.11400    0.01955  -5.831 6.57e-09 ***
## Gulf_Coast.l2     -0.11428    0.04097  -2.789 0.005341 **
## Rocky_Mountain.l2 -0.08354    0.02401  -3.479 0.000516 ***
## West_Coast.l2     -0.56639    0.02561 -22.119  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
```

```
## Residual standard error: 0.01327 on 1710 degrees of freedom
## Multiple R-Squared: 0.9992,  Adjusted R-squared: 0.9992
## F-statistic: 2.069e+05 on 10 and 1710 DF,  p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##                East_Coast   Midwest Gulf_Coast Rocky_Mountain West_Coast
## East_Coast      0.0002115 0.0002572  0.0002262      1.162e-04  1.081e-04
## Midwest         0.0002572 0.0005990  0.0003182      1.767e-04  1.562e-04
## Gulf_Coast      0.0002262 0.0003182  0.0003070      1.337e-04  1.243e-04
## Rocky_Mountain  0.0001162 0.0001767  0.0001337      2.256e-04  8.621e-05
## West_Coast      0.0001081 0.0001562  0.0001243      8.621e-05  1.759e-04
##
## Correlation matrix of residuals:
##                East_Coast Midwest Gulf_Coast Rocky_Mountain West_Coast
## East_Coast         1.0000  0.7227     0.8877         0.5322     0.5603
## Midwest            0.7227  1.0000     0.7420         0.4808     0.4810
## Gulf_Coast         0.8877  0.7420     1.0000         0.5082     0.5349
## Rocky_Mountain     0.5322  0.4808     0.5082         1.0000     0.4327
## West_Coast         0.5603  0.4810     0.5349         0.4327     1.0000
```

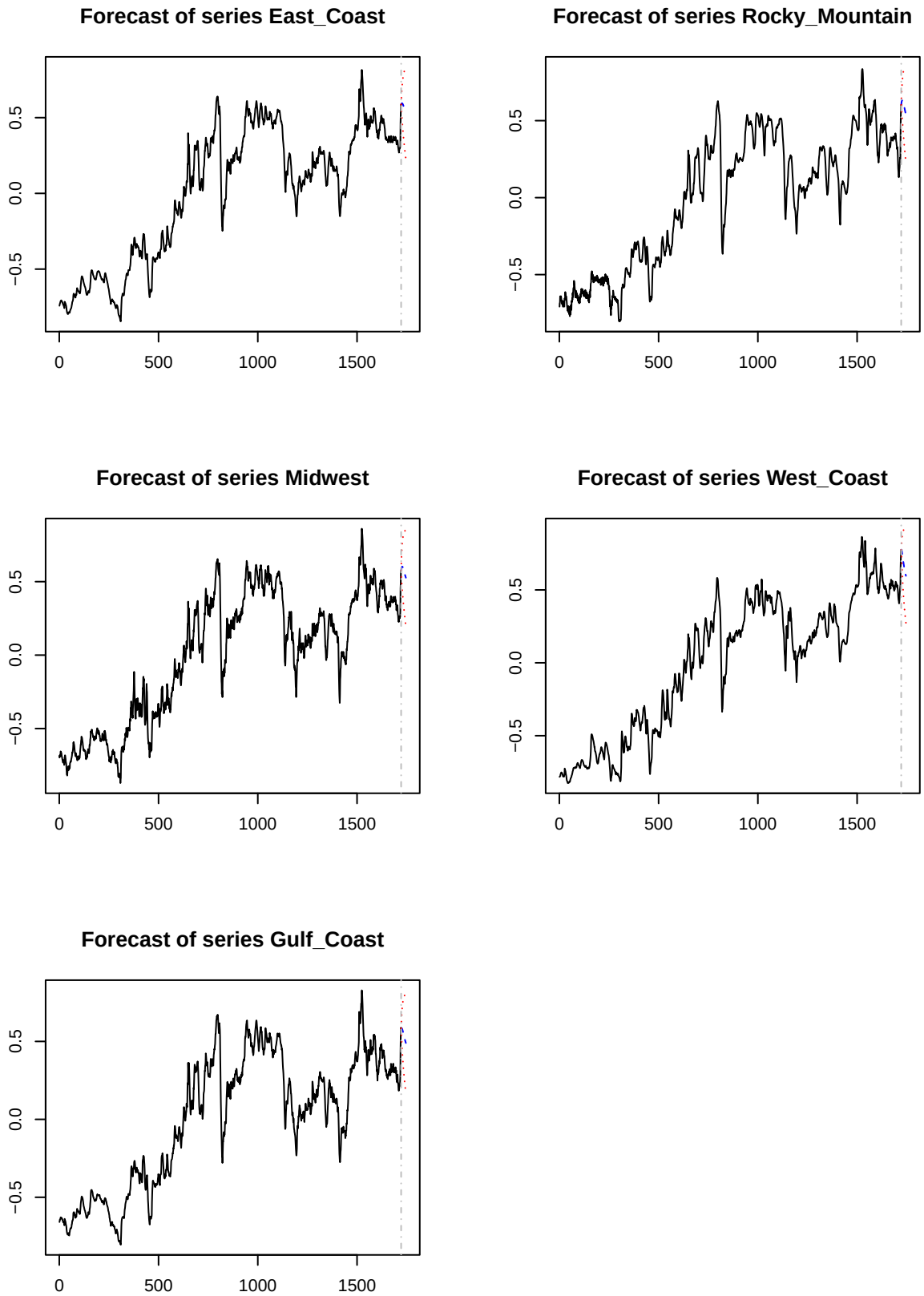
```r
plot(ts(resid(fit_regional), start = c(1993, 14), frequency = 52))
```



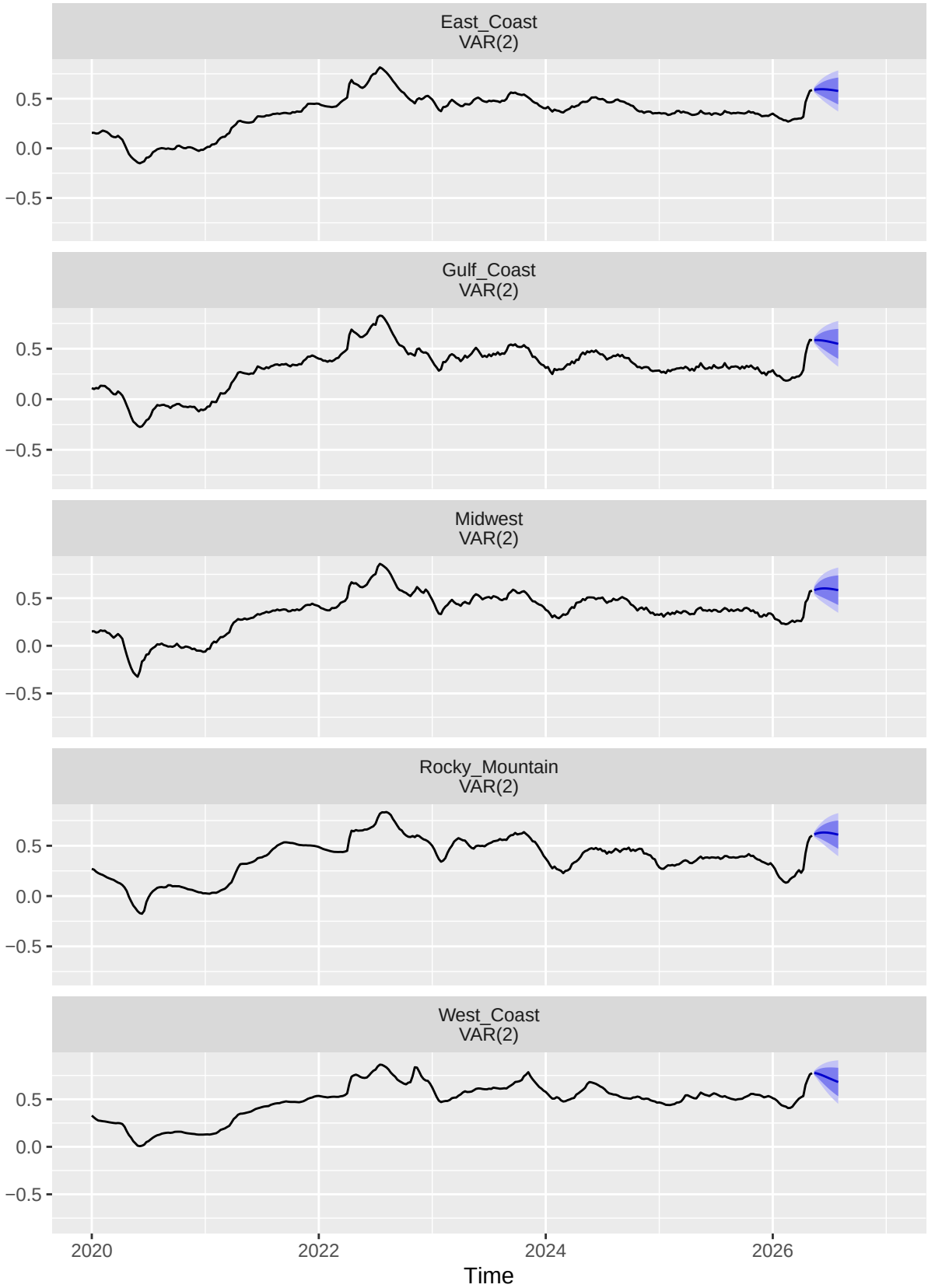

```r
acf(resid(fit_regional), 52)
```


East_Coast
Es_C & Mdws
Es_C & Gl_C
Es_C & Rc_M
Es_C & Ws_C
Mdws & Es_C
Midwest
Mdws & Gl_C
Mdws & Rc_M
Mdws & Ws_C
Gl_C & Es_C
Gl_C & Mdws
Gulf_Coast
Gl_C & Rc_M
Gl_C & Ws_C
Rc_M & Es_C
Rc_M & Mdws
Rc_M & Gl_C
Rocky_Mountain
Rc_M & Ws_C
Ws_C & Es_C
Ws_C & Mdws
Ws_C & Gl_C
Ws_C & Rc_M
West_Coast
ACF
0.0
0.4
0.8
Lag
−50
−30
−10
0
20
40

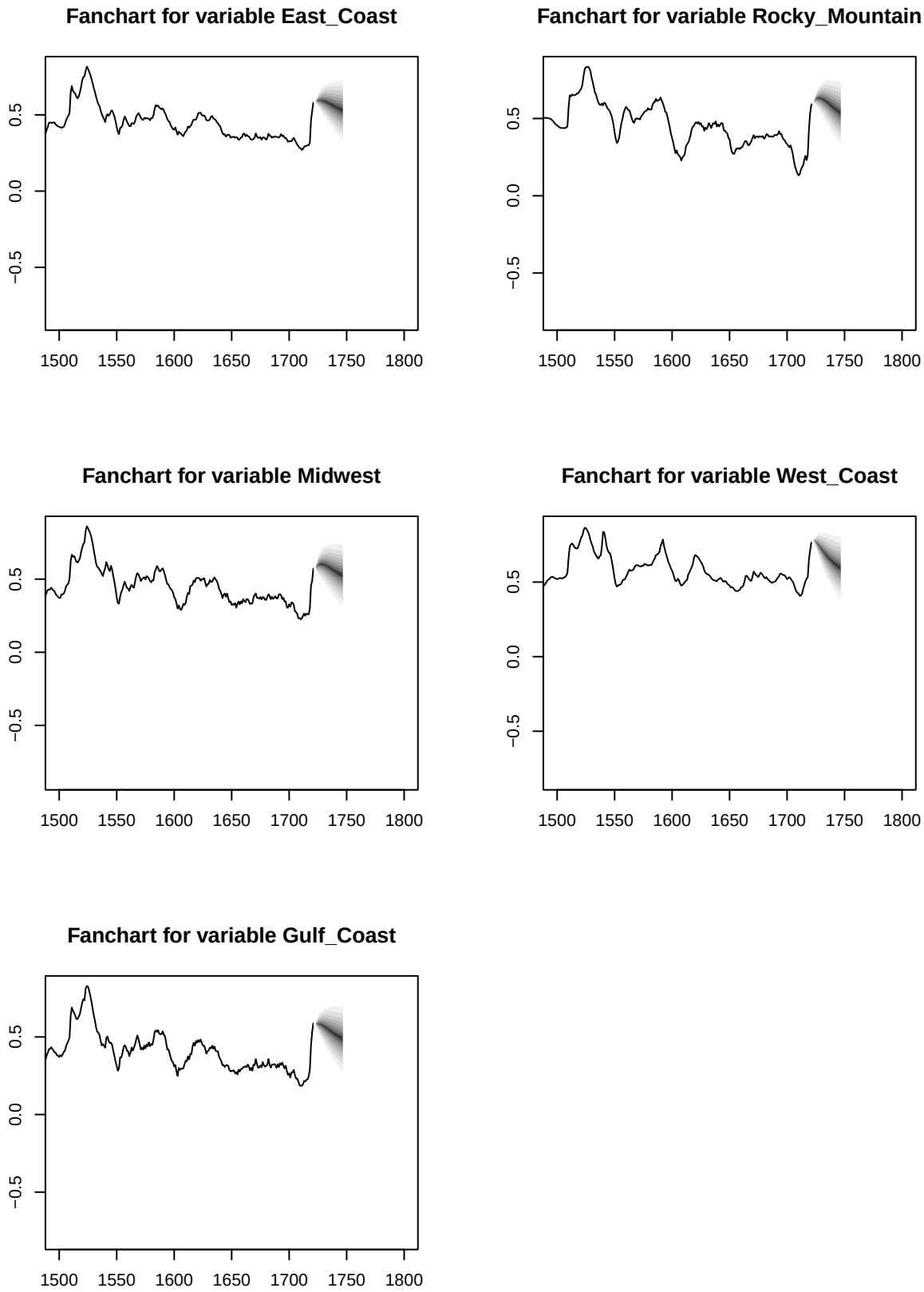
```r
pred_regional = predict(fit_regional, n.ahead = 24, ci = 0.95)
plot(pred_regional)
```


Forecast of series East_Coast
Forecast of series Rocky_Mountain
Forecast of series Midwest
Forecast of series West_Coast
Forecast of series Gulf_Coast
0.5
0.0
−0.5
0
500
1000
1500

```r
forecast(fit_regional, h = 12) |>
  autoplot() + xlim(c(2020, 2027))
```


East_Coast
VAR(2)
Gulf_Coast
VAR(2)
Midwest
VAR(2)
Rocky_Mountain
VAR(2)
West_Coast
VAR(2)
0.5
0.0
−0.5
2020
2022
2024
2026
Time

```r
fanchart(pred_regional, xlim = c(1500,1800))
```


Fanchart for variable East_Coast
Fanchart for variable Rocky_Mountain
Fanchart for variable Midwest
Fanchart for variable West_Coast
Fanchart for variable Gulf_Coast
0.5
0.0
−0.5
1500
1550
1600
1650
1700
1750
1800

Model Selection - Crude + US Retail VAR Model

```r
# log for variation
gas_crude$US_All_Grades <- log(gas_crude$US_All_Grades)
gas_crude$Brent <- log(gas_crude$Brent)

# subtract mean to detrend
gas_crude$US_All_Grades <- gas_crude$US_All_Grades - mean(gas_crude$US_All_Grades)
gas_crude$Brent <- gas_crude$Brent - mean(gas_crude$Brent)
```

```r
x = ts(dplyr::select(gas_crude, US_All_Grades, Brent),
       # fourteenth week of 1993
       start = c(1993, 14), frequency = 52)
VARselect(x, lag.max = 14, type = 'none')
```

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      6      4      3      6
##
## $criteria
##                    1             2             3             4             5
## AIC(n) -1.444751e+01 -1.492898e+01 -1.494558e+01 -1.495174e+01 -1.495467e+01
## HQ(n)  -1.444279e+01 -1.491954e+01 -1.493142e+01 -1.493286e+01 -1.493106e+01
## SC(n)  -1.443475e+01 -1.490347e+01 -1.490732e+01 -1.490072e+01 -1.489090e+01
## FPE(n)  5.315297e-07  3.284176e-07  3.230096e-07  3.210279e-07  3.200893e-07
##                    6             7             8             9            10
## AIC(n) -1.495870e+01 -1.495470e+01 -1.495262e+01 -1.494953e+01 -1.494977e+01
## HQ(n)  -1.493038e+01 -1.492166e+01 -1.491485e+01 -1.490705e+01 -1.490257e+01
## SC(n)  -1.488218e+01 -1.486543e+01 -1.485059e+01 -1.483475e+01 -1.482224e+01
## FPE(n)  3.188000e-07  3.200788e-07  3.207462e-07  3.217384e-07  3.216603e-07
##                   11            12            13            14
## AIC(n) -1.494958e+01 -1.494737e+01 -1.494300e+01 -1.493944e+01
## HQ(n)  -1.489765e+01 -1.489072e+01 -1.488163e+01 -1.487335e+01
## SC(n)  -1.480929e+01 -1.479433e+01 -1.477720e+01 -1.476089e+01
## FPE(n)  3.217231e-07  3.224349e-07  3.238477e-07  3.250036e-07
```

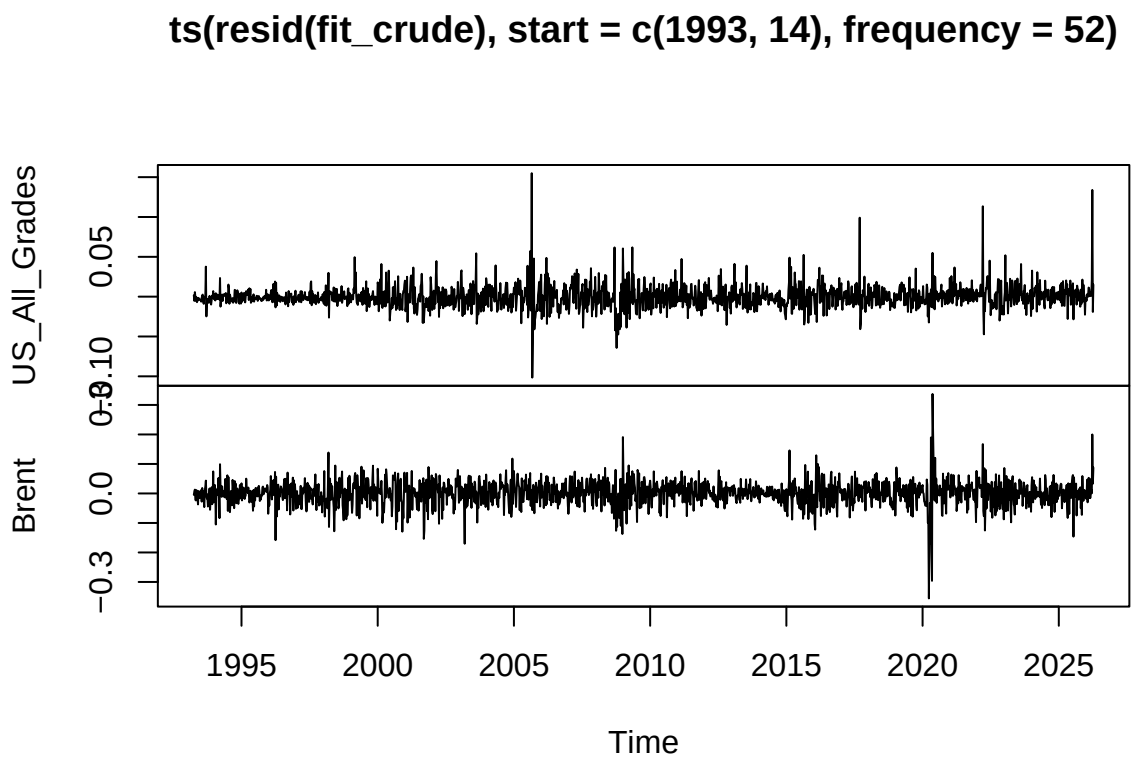
Using p = 3 due to SC (same as BIC).

```r
fit_crude = VAR(x, p = 3, type = 'none')
summary(fit_crude)
```

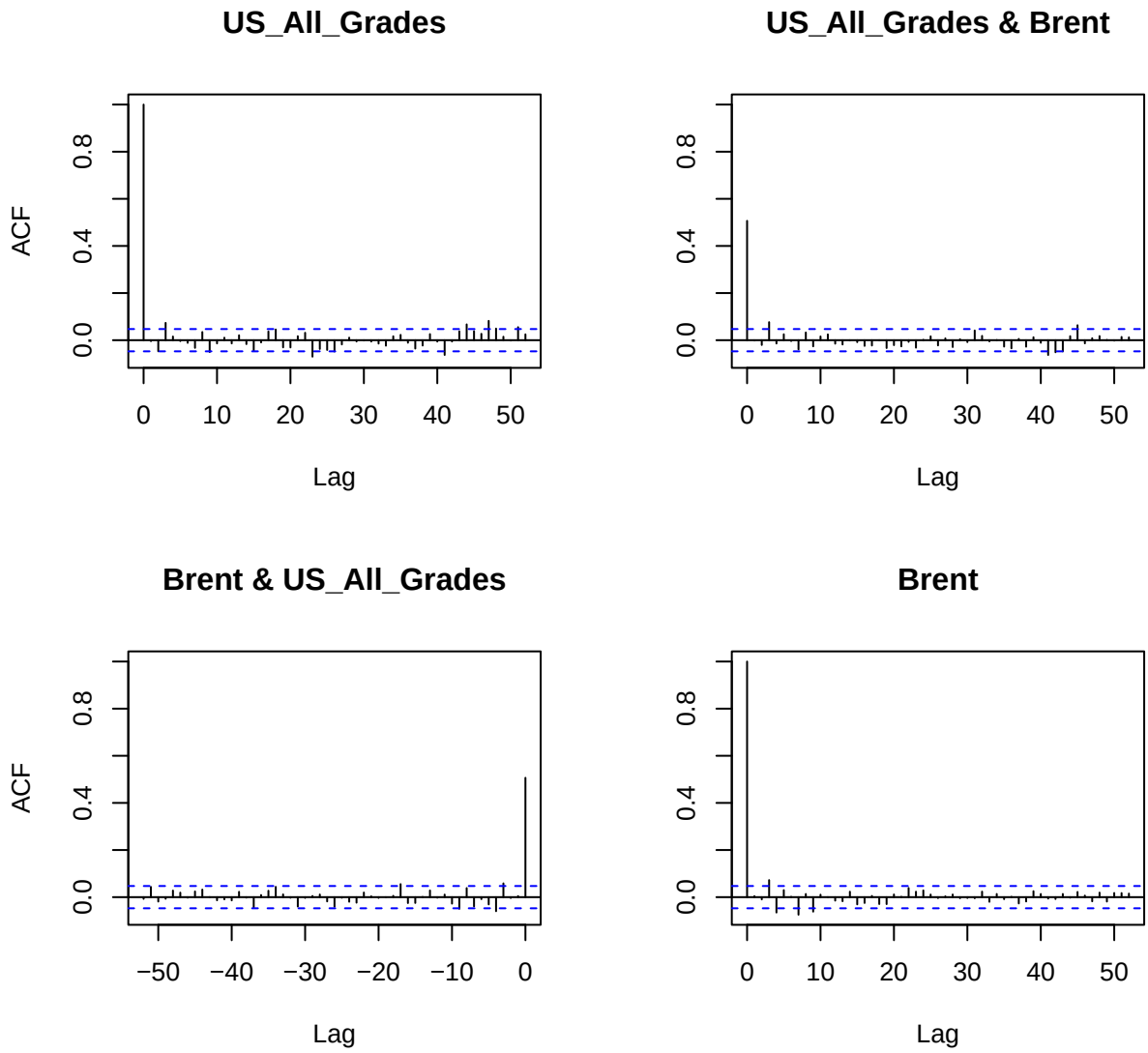
```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: US_All_Grades, Brent
## Deterministic variables: none
## Sample size: 1718
## Log Likelihood: 7984.64
## Roots of the characteristic polynomial:
## 0.9961 0.9469 0.6831 0.2265 0.2265 0.1045
## Call:
## VAR(y = x, p = 3, type = "none")
##
##
## Estimation results for equation US_All_Grades:
```

```
## ==============================================
## US_All_Grades = US_All_Grades.l1 + Brent.l1 + US_All_Grades.l2 + Brent.l2 + US_All_Grades.l3 + Brent
##
##                    Estimate Std. Error t value Pr(>|t|)
## US_All_Grades.l1  1.394158   0.027881  50.004  < 2e-16 ***
## Brent.l1          0.098471   0.009757  10.092  < 2e-16 ***
## US_All_Grades.l2 -0.264919   0.046108  -5.746 1.08e-08 ***
## Brent.l2         -0.117040   0.014426  -8.113 9.33e-16 ***
## US_All_Grades.l3 -0.147515   0.025697  -5.741 1.11e-08 ***
## Brent.l3          0.029594   0.009807   3.018  0.00259 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.01508 on 1712 degrees of freedom
## Multiple R-Squared: 0.9988,  Adjusted R-squared: 0.9988
## F-statistic: 2.387e+05 on 6 and 1712 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation Brent:
## ======================================
## Brent = US_All_Grades.l1 + Brent.l1 + US_All_Grades.l2 + Brent.l2 + US_All_Grades.l3 + Brent.l3
##
##                   Estimate Std. Error t value Pr(>|t|)
## US_All_Grades.l1  0.03180    0.08011   0.397  0.69142
## Brent.l1          1.24560    0.02804  44.430  < 2e-16 ***
## US_All_Grades.l2  0.28518    0.13248   2.153  0.03149 *
## Brent.l2         -0.34697    0.04145  -8.370  < 2e-16 ***
## US_All_Grades.l3 -0.29060    0.07384  -3.936 8.62e-05 ***
## Brent.l3          0.08173    0.02818   2.900  0.00377 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.04332 on 1712 degrees of freedom
## Multiple R-Squared: 0.9959,  Adjusted R-squared: 0.9959
## F-statistic: 6.928e+04 on 6 and 1712 DF,  p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##               US_All_Grades     Brent
## US_All_Grades     0.0002272 0.0003307
## Brent             0.0003307 0.0018763
##
## Correlation matrix of residuals:
##               US_All_Grades  Brent
## US_All_Grades        1.0000 0.5065
## Brent                0.5065 1.0000
```

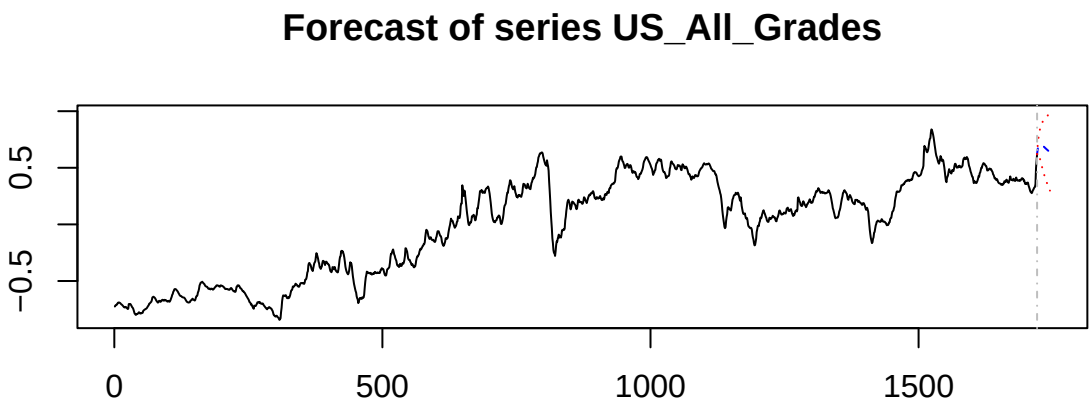
```r
plot(ts(resid(fit_crude), start = c(1993, 14), frequency = 52))
```

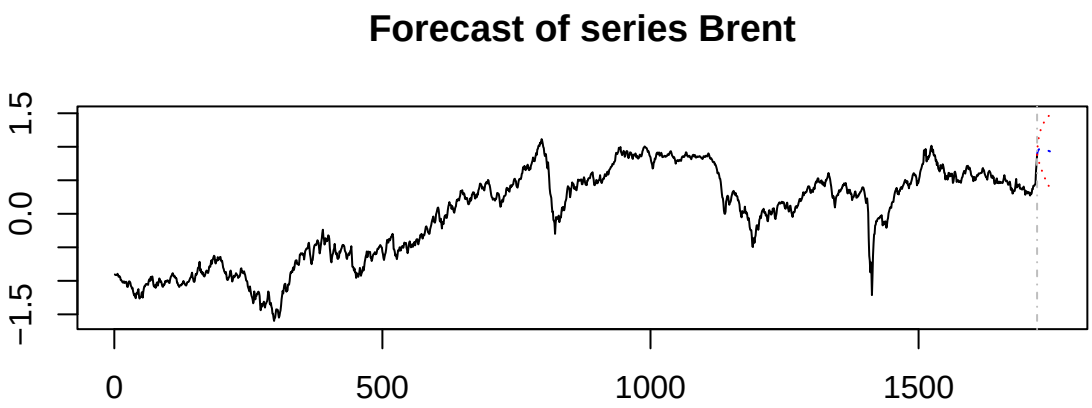



```r
acf(resid(fit_crude), 52)
```

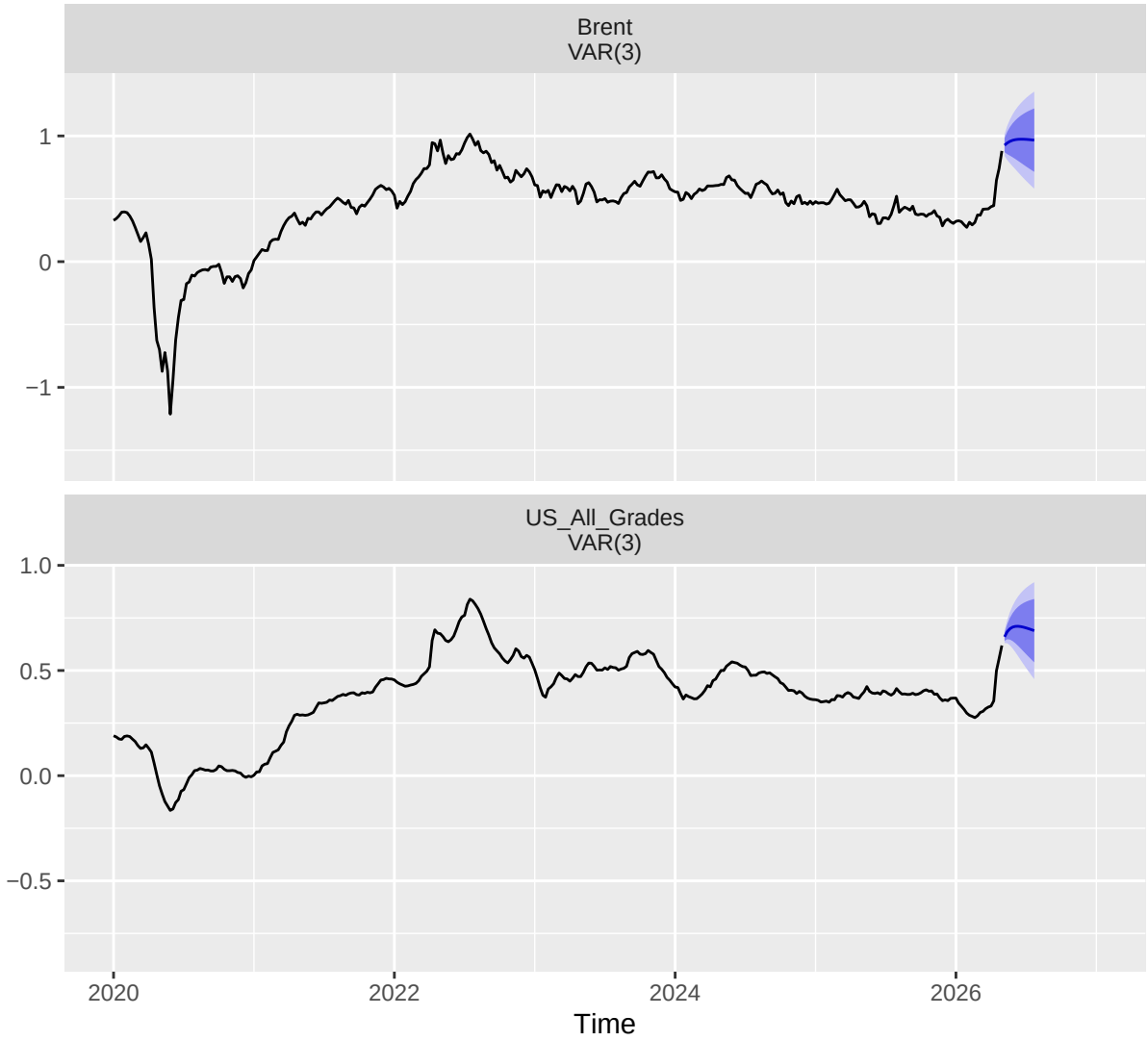



```r
pred_crude = predict(fit_crude, n.ahead = 24, ci = 0.95)
plot(pred_crude)
```

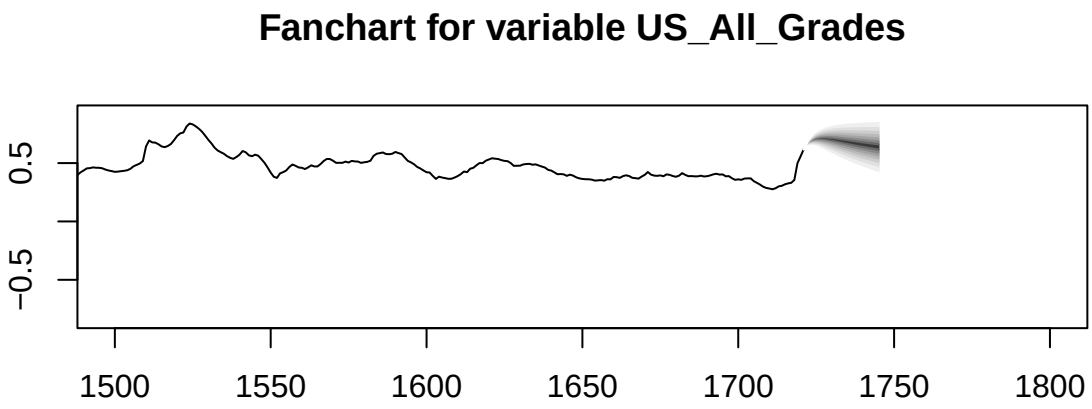


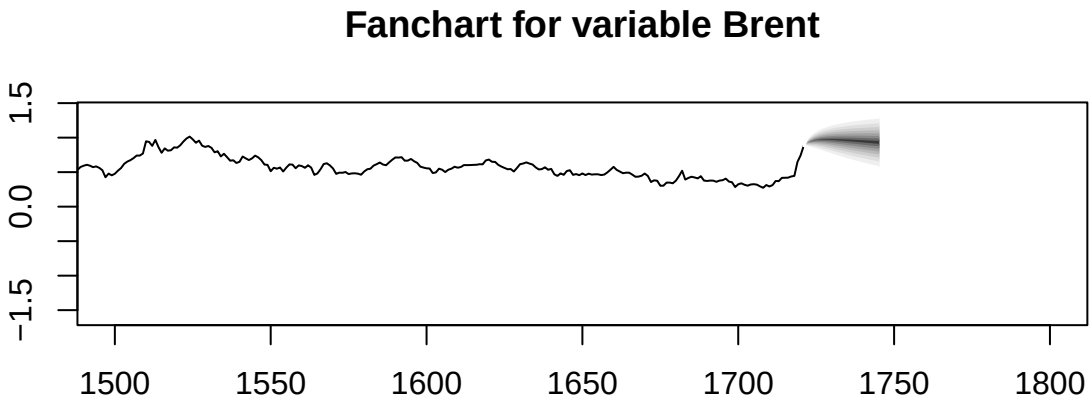



```
forecast(fit_crude, h = 12) |>
  autoplot() + xlim(c(2020, 2027))
```




```
fanchart(pred_crude, xlim = c(1500,1800))
```


Fanchart for variable US_All_Grades
0.5
−0.5
1500
1550
1600
1650
1700
1750
1800


Fanchart for variable Brent
1.5
0.0
−1.5
1500
1550
1600
1650
1700
1750
1800